KNHANES 2020–2024 (pooled), 5 waves
N = 34,640 respondents assessed
Candidate variables: 66 assessed → 57 retained

Variables excluded (n = 9)
• Missingness, NaN > 30% (n=5):
DI4_dg, DE1_4, BS3_1,
BD2_1, LW_ms
• Redundancy / definitional
overlap, a priori (n=4):
HE_obe, HE_wc, HE_glu, N_EN

Age-eligibility filter
A (≥50): n = 17,545
B (≥65): n = 9,003

Excluded, below age threshold
A (age < 50): excl. 17,095
B (age < 65): excl. 25,637

Missing/invalid target (O_TC) excluded
A: analytic cohort n = 14,871
B: analytic cohort n = 7,704

Excluded, missing/invalid O_TC
A: excl. 2,674
B: excl. 1,299

Data preparation
• Missingness/collinearity screen (VIF): confirms
the 9 a priori exclusions above; 0 further
variables dropped from the 57 retained
• Derive eGFR, diabetes, hypertension, brush_freq
• Missing predictor values imputed (mode/median,
fit on development set only)

ML (predictive) arm
Temporal split:
Development (2020–2023) vs.
Evaluation (2024)

Inferential (confirmatory) arm
Full pooled cohort,
survey-weighted (strata/PSU/weight)
No development/evaluation split

A
Dev n = 11,536
Eval n = 3,335

B
Dev n = 5,946
Eval n = 1,758

A
n = 14,871

B
n = 7,704